Proofs of Theorem 3.2 and Theorem 3.3

A Proofs for Pooling-Induced Representation Contraction

This text provides the complete derivations for Theorem 3.2 and Theorem 3.3 in main paper. We first establish the finite-population sampling variance shared by both proofs.

A.1 Finite-Population Sampling Variance

Recall the pooling construction

$$u_i = \frac{1}{k+1}\left(v_i + \sum_{x \in S_i} x\right),$$

where S_i contains k vectors sampled uniformly without replacement from $V \setminus \{v_i\}$.

Define

$$\bar{v}_{-i} = \frac{1}{n-1}\sum_{j \neq i} v_j = \frac{n\mu - v_i}{n-1}. \tag{1}$$

From uniform sampling,

$$\begin{aligned}\mathbb{E}\left[\sum_{x \in S_i} x\right] &= k\bar{v}_{-i} \\ &= \frac{k}{n-1}\left(n\mu - v_i\right),\end{aligned} \tag{2}$$

and therefore

$$\begin{aligned}\mathbb{E}[u_i] &= \frac{1}{k+1}\left[v_i + \frac{k}{n-1}\left(n\mu - v_i\right)\right] \\ &= \alpha(k)v_i + (1-\alpha(k))\,\mu \\ &= \mu + \alpha(k)(v_i - \mu),\end{aligned} \tag{3}$$

where

$$\alpha(k) = \frac{n-1-k}{(k+1)(n-1)}.$$

Define the sampling fluctuation

$$\varepsilon_i = u_i - \mathbb{E}[u_i]. \tag{4}$$

Then

$$u_i - \mu = \alpha(k)(v_i - \mu) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i] = 0. \tag{5}$$

Lemma A.1 (Finite-Population Pooling Variance). *For every i,*

$$\mathbb{E}\|\varepsilon_i\|_2^2 = \frac{k(n-1-k)}{(k+1)^2(n-2)}\left(1 - \|\bar{v}_{-i}\|_2^2\right). \tag{6}$$

Consequently,

$$\mathbb{E}\|\varepsilon_i\|_2^2 \leq \frac{k(n-1-k)}{(k+1)^2(n-2)}. \tag{7}$$

Averaging over all i further gives

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\|\varepsilon_i\|_2^2 = \frac{nk(n-1-k)}{(k+1)^2(n-1)^2}\left(1 - \|\mu\|_2^2\right). \tag{8}$$

Proof. Fix i and let

$$N = n - 1.$$

Enumerate the vectors in $V \setminus \{v_i\}$ as

$$x_1, \ldots, x_N,$$

and let their population mean be

$$\bar{x} = \frac{1}{N}\sum_{r=1}^{N} x_r = \bar{v}_{-i}.$$

For every r, let $I_r \in \{0,1\}$ indicate whether x_r is included in S_i. Since exactly k elements are sampled uniformly without replacement,

$$\mathbb{E}[I_r] = \frac{k}{N}. \tag{9}$$

Because I_r is Bernoulli with success probability k/N,

$$\begin{aligned}\mathrm{Var}(I_r) &= \frac{k}{N}\left(1 - \frac{k}{N}\right) \\ &= \frac{k(N-k)}{N^2}.\end{aligned} \tag{10}$$

For $r \neq s$,

$$\mathbb{P}(I_r = 1, I_s = 1) = \frac{k(k-1)}{N(N-1)}.$$

Hence

$$\begin{aligned}\mathrm{Cov}(I_r, I_s) &= \mathbb{E}[I_r I_s] - \mathbb{E}[I_r]\mathbb{E}[I_s] \\ &= \frac{k(k-1)}{N(N-1)} - \frac{k^2}{N^2} \\ &= -\frac{k(N-k)}{N^2(N-1)}.\end{aligned} \tag{11}$$

Define the centered population vectors

$$y_r = x_r - \bar{x}.$$

Then

$$\sum_{r=1}^{N} y_r = 0.$$

Furthermore,

$$\begin{aligned}\sum_{x \in S_i} x - k\bar{x} &= \sum_{r=1}^{N} I_r x_r - k\bar{x} \\ &= \sum_{r=1}^{N} I_r (x_r - \bar{x}) \\ &= \sum_{r=1}^{N} I_r y_r,\end{aligned} \tag{12}$$

where we used

$$\sum_{r=1}^{N} I_r = k.$$

Since

$$\mathbb{E}\left[\sum_{r=1}^{N} I_r y_r\right] = \frac{k}{N}\sum_{r=1}^{N} y_r = 0,$$

we obtain

$$\begin{aligned}\mathbb{E}\left\|\sum_{r=1}^{N} I_r y_r\right\|_2^2 &= \sum_{r=1}^{N}\mathrm{Var}(I_r)\|y_r\|_2^2 \\ &\quad + \sum_{r \neq s}\mathrm{Cov}(I_r, I_s) y_r^\top y_s.\end{aligned} \tag{13}$$

Because

$$\sum_{r=1}^{N} y_r = 0,$$

we also have

$$\begin{aligned} 0 &= \left\| \sum_{r=1}^{N} y_r \right\|_2^2 \\ &= \sum_{r=1}^{N} \|y_r\|_2^2 + \sum_{r \neq s} y_r^\top y_s, \end{aligned} \tag{14}$$

so

$$\sum_{r \neq s} y_r^\top y_s = -\sum_{r=1}^{N} \|y_r\|_2^2. \tag{15}$$

Substituting Equation 10, Equation 11, and Equation 15 into Equation 13,

$$\begin{aligned} &\mathbb{E}\left\| \sum_{x \in S_i} x - k\bar{x} \right\|_2^2 \\ &= \left[\frac{k(N-k)}{N^2} + \frac{k(N-k)}{N^2(N-1)} \right] \sum_{r=1}^{N} \|y_r\|_2^2 \\ &= \frac{k(N-k)}{N(N-1)} \sum_{r=1}^{N} \|x_r - \bar{x}\|_2^2. \end{aligned} \tag{16}$$

From Equation 4,

$$\varepsilon_i = \frac{1}{k+1} \left(\sum_{x \in S_i} x - k\bar{v}_{-i} \right). \tag{17}$$

Therefore,

$$\begin{aligned} \mathbb{E}\|\varepsilon_i\|_2^2 &= \frac{k(N-k)}{(k+1)^2 N(N-1)} \sum_{r=1}^{N} \|x_r - \bar{x}\|_2^2 \\ &= \frac{k(N-k)}{(k+1)^2(N-1)} \left[\frac{1}{N} \sum_{r=1}^{N} \|x_r - \bar{x}\|_2^2 \right]. \end{aligned} \tag{18}$$

Since every x_r is a unit vector,

$$\begin{aligned} \frac{1}{N} \sum_{r=1}^{N} \|x_r - \bar{x}\|_2^2 &= \frac{1}{N} \sum_{r=1}^{N} \left(\|x_r\|_2^2 - 2x_r^\top \bar{x} + \|\bar{x}\|_2^2 \right) \\ &= 1 - 2\bar{x}^\top \left(\frac{1}{N} \sum_{r=1}^{N} x_r \right) + \|\bar{x}\|_2^2 \\ &= 1 - \|\bar{x}\|_2^2. \end{aligned} \tag{19}$$

Restoring $N = n - 1$ and $\bar{x} = \bar{v}_{-i}$ gives

$$\mathbb{E}\|\varepsilon_i\|_2^2 = \frac{k(n-1-k)}{(k+1)^2(n-2)} \left(1 - \|\bar{v}_{-i}\|_2^2 \right),$$

which proves Equation 6. Since $\|\bar{v}_{-i}\|_2^2 \geq 0$, the upper bound Equation 7 follows immediately.

It remains to average over i. From

$$\bar{v}_{-i} = \frac{n\mu - v_i}{n-1},$$

we obtain

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^{n} \|\bar{v}_{-i}\|_2^2 &= \frac{1}{(n-1)^2} \frac{1}{n} \sum_{i=1}^{n} \|n\mu - v_i\|_2^2 \\ &= \frac{1}{(n-1)^2} \frac{1}{n} \sum_{i=1}^{n} \left(n^2 \|\mu\|_2^2 - 2n\mu^\top v_i + 1 \right). \end{aligned} \tag{20}$$

Because

$$\frac{1}{n} \sum_{i=1}^{n} \mu^\top v_i = \mu^\top \left(\frac{1}{n} \sum_{i=1}^{n} v_i \right) = \|\mu\|_2^2,$$

we obtain

$$\frac{1}{n} \sum_{i=1}^{n} \|\bar{v}_{-i}\|_2^2 = \frac{n(n-2)\|\mu\|_2^2 + 1}{(n-1)^2}. \tag{21}$$

Hence

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^{n} \left(1 - \|\bar{v}_{-i}\|_2^2 \right) &= 1 - \frac{n(n-2)\|\mu\|_2^2 + 1}{(n-1)^2} \\ &= \frac{n(n-2)}{(n-1)^2} \left(1 - \|\mu\|_2^2 \right). \end{aligned} \tag{22}$$

Averaging Equation 6 and applying Equation 22 gives

$$\frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\|\varepsilon_i\|_2^2 = \frac{nk(n-1-k)}{(k+1)^2(n-1)^2} \left(1 - \|\mu\|_2^2 \right),$$

proving Equation 8. □

A.2 Proof of Theorem (Pooling-Induced Unnormalized Contraction)

Theorem 3.2 (Pooling-Induced Unnormalized Contraction)

Proof. We first consider the norm-sensitive dot-product statistic.

From Equation 3,

$$\mathbb{E}[u_i] = \alpha v_i + (1-\alpha)\mu,$$

where we suppress the dependence of α on k for readability.

Since the sampling procedures for S_i and S_j are independent when $i \neq j$,

$$\mathbb{E}[u_i^\top u_j] = \mathbb{E}[u_i]^\top \mathbb{E}[u_j]. \tag{23}$$

Therefore,

$$\begin{aligned} \mathbb{E}[u_i^\top u_j] &= \left[\alpha v_i + (1-\alpha)\mu \right]^\top \left[\alpha v_j + (1-\alpha)\mu \right] \\ &= \alpha^2 v_i^\top v_j + \alpha(1-\alpha) \left(v_i^\top \mu + v_j^\top \mu \right) + (1-\alpha)^2 \|\mu\|_2^2. \end{aligned} \tag{24}$$

Define

$$S_U = \frac{1}{n(n-1)} \sum_{i \neq j} u_i^\top u_j.$$

Averaging Equation 24 over all ordered pairs $i \neq j$, the first term becomes $\alpha^2 S_V$.

For the cross term,

$$\begin{aligned} \frac{1}{n(n-1)} \sum_{i \neq j} v_i^\top \mu &= \frac{1}{n(n-1)} (n-1) \sum_{i=1}^{n} v_i^\top \mu \\ &= \frac{1}{n} \left(\sum_{i=1}^{n} v_i \right)^\top \mu \\ &= \|\mu\|_2^2. \end{aligned} \tag{25}$$

By symmetry, the same result holds for $v_j^\top \mu$. Hence

$$\begin{aligned}\mathbb{E}[S_U] &= \alpha^2 S_V + 2\alpha(1-\alpha)\|\mu\|_2^2 + (1-\alpha)^2\|\mu\|_2^2 \\ &= \alpha^2 S_V + (1-\alpha^2)\|\mu\|_2^2. \end{aligned} \tag{26}$$

The global mean satisfies

$$\begin{aligned}\|\mu\|_2^2 &= \left\| \frac{1}{n}\sum_{i=1}^n v_i \right\|_2^2 \\ &= \frac{1}{n^2}\left(\sum_{i=1}^n \|v_i\|_2^2 + \sum_{i\neq j} v_i^\top v_j \right) \\ &= \frac{n + n(n-1)S_V}{n^2} \\ &= \frac{1 + (n-1)S_V}{n}. \end{aligned} \tag{27}$$

Since

$$\mathrm{MPD}_{\mathrm{dot}}(V) = 1 - S_V$$

and

$$\mathbb{E}[\mathrm{MPD}_{\mathrm{dot}}(U)] = 1 - \mathbb{E}[S_U],$$

we obtain

$$\begin{aligned}&\mathrm{MPD}_{\mathrm{dot}}(V) - \mathbb{E}[\mathrm{MPD}_{\mathrm{dot}}(U)] \\ &= \mathbb{E}[S_U] - S_V \\ &= (1-\alpha^2)\left(\|\mu\|_2^2 - S_V\right) \\ &= (1-\alpha^2)\left[\frac{1 + (n-1)S_V}{n} - S_V\right] \\ &= (1-\alpha^2)\frac{1 - S_V}{n}. \end{aligned} \tag{28}$$

Because $S_V < 1$ and $0 \le \alpha(k) < 1$ for every $k \ge 1$, this reduction is strictly positive.

Furthermore,

$$\alpha(k) = \frac{n-1-k}{(k+1)(n-1)}$$

strictly decreases as the integer k increases. Hence $1-\alpha(k)^2$ strictly increases, proving the scale dependence in the first part of the theorem.

We next consider squared-Euclidean dispersion. From Equation 5,

$$u_i - \mu = \alpha(v_i - \mu) + \varepsilon_i.$$

Thus,

$$u_i - u_j = \alpha(v_i - v_j) + \varepsilon_i - \varepsilon_j.$$

For $i \neq j$, ε_i and ε_j are independent and zero mean. Therefore,

$$\begin{aligned}\mathbb{E}\|u_i - u_j\|_2^2 &= \alpha^2\|v_i - v_j\|_2^2 + \mathbb{E}\|\varepsilon_i - \varepsilon_j\|_2^2 \\ &= \alpha^2\|v_i - v_j\|_2^2 + \mathbb{E}\|\varepsilon_i\|_2^2 + \mathbb{E}\|\varepsilon_j\|_2^2. \end{aligned} \tag{29}$$

Averaging over all ordered pairs,

$$\begin{aligned}\mathbb{E}\left[\mathrm{MPD}_{\mathrm{Euc}}(U)\right] &= \frac{1}{n(n-1)}\sum_{i\neq j}\mathbb{E}\|u_i - u_j\|_2^2 \\ &= \alpha^2\mathrm{MPD}_{\mathrm{Euc}}(V) + \frac{2}{n}\sum_{i=1}^n \mathbb{E}\|\varepsilon_i\|_2^2. \end{aligned} \tag{30}$$

From Equation 27,

$$\begin{aligned}1 - \|\mu\|_2^2 &= 1 - \frac{1 + (n-1)S_V}{n} \\ &= \frac{n-1}{n}(1 - S_V). \end{aligned} \tag{31}$$

Because the original vectors are unit normalized,

$$\mathrm{MPD}_{\mathrm{Euc}}(V) = 2(1 - S_V),$$

and hence

$$1 - \|\mu\|_2^2 = \frac{n-1}{2n}\mathrm{MPD}_{\mathrm{Euc}}(V). \tag{32}$$

By Lemma A.1,

$$\frac{1}{n}\sum_{i=1}^n \mathbb{E}\|\varepsilon_i\|_2^2 = \frac{nk(n-1-k)}{(k+1)^2(n-1)^2}\left(1 - \|\mu\|_2^2\right).$$

Substituting this identity and Equation 32 into Equation 30,

$$\begin{aligned}\mathbb{E}[\mathrm{MPD}_{\mathrm{Euc}}(U)] &= \alpha^2\mathrm{MPD}_{\mathrm{Euc}}(V) \\ &\quad + 2\frac{nk(n-1-k)}{(k+1)^2(n-1)^2}\left(1 - \|\mu\|_2^2\right) \\ &= \left[\alpha^2 + \frac{k(n-1-k)}{(k+1)^2(n-1)}\right]\mathrm{MPD}_{\mathrm{Euc}}(V). \end{aligned} \tag{33}$$

We now simplify the coefficient. Since

$$\alpha = \frac{n-1-k}{(k+1)(n-1)},$$

we have

$$\alpha^2 + \frac{k(n-1-k)}{(k+1)^2(n-1)} = \frac{(n-1)\alpha + \alpha^2}{n}. \tag{34}$$

Thus

$$\mathbb{E}[\mathrm{MPD}_{\mathrm{Euc}}(U)] = r(k)\mathrm{MPD}_{\mathrm{Euc}}(V),$$

where

$$r(k) = \frac{(n-1)\alpha(k) + \alpha(k)^2}{n}.$$

To establish monotonicity, define

$$f(\alpha) = \frac{(n-1)\alpha + \alpha^2}{n}.$$

Then

$$\frac{df}{d\alpha} = \frac{n-1+2\alpha}{n} > 0$$

for $\alpha \ge 0$. Since $\alpha(k)$ strictly decreases with k, $r(k) = f(\alpha(k))$ also strictly decreases.

For every $k \ge 1$,

$$0 \le \alpha(k) < 1.$$

Because f is increasing,

$$0 \le r(k) < f(1) = 1.$$

Finally,

$$k = n-1 \quad\Longrightarrow\quad \alpha(k) = 0 \quad\Longrightarrow\quad r(k) = 0.$$

At this global pooling scale,

$$u_i = \frac{1}{n}\sum_{j=1}^n v_j = \mu$$

for every i, so all pairwise Euclidean distances vanish. □

A.3 Proof of Theorem (Angular Concentration under a Nonzero Global Mean)

Theorem 3.3 (Angular Concentration under a Nonzero Global Mean).

Proof. From Equation 5,

$$u_i - \mu = \alpha(k)(v_i - \mu) + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i] = 0. \tag{35}$$

We first establish the mean-square concentration identity. Taking squared norms and expectations,

$$\mathbb{E}\|u_i - \mu\|_2^2 = \alpha(k)^2\|v_i - \mu\|_2^2 + \mathbb{E}\|\varepsilon_i\|_2^2, \tag{36}$$

because the cross term vanishes:

$$\mathbb{E}\left[(v_i - \mu)^\top \varepsilon_i\right] = (v_i - \mu)^\top \mathbb{E}[\varepsilon_i] = 0.$$

The deterministic component satisfies

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^n \|v_i - \mu\|_2^2 &= \frac{1}{n}\sum_{i=1}^n \left(\|v_i\|_2^2 - 2v_i^\top \mu + \|\mu\|_2^2\right) \\ &= 1 - 2\mu^\top\left(\frac{1}{n}\sum_{i=1}^n v_i\right) + \|\mu\|_2^2 \\ &= 1 - \|\mu\|_2^2. \end{aligned} \tag{37}$$

Averaging Equation 36 over i and applying Lemma A.1,

$$\frac{1}{n}\sum_{i=1}^n \mathbb{E}\|u_i - \mu\|_2^2 = \left[\alpha(k)^2 + \frac{nk(n-1-k)}{(k+1)^2(n-1)^2}\right]\left(1 - \|\mu\|_2^2\right). \tag{38}$$

Let

$$q = n - 1 - k.$$

Since

$$\alpha(k) = \frac{q}{(k+1)(n-1)},$$

the coefficient becomes

$$\begin{aligned}\alpha(k)^2 + \frac{nk(n-1-k)}{(k+1)^2(n-1)^2} &= \frac{q^2 + nkq}{(k+1)^2(n-1)^2} \\ &= \frac{q(q+nk)}{(k+1)^2(n-1)^2}. \end{aligned} \tag{39}$$

Because

$$q + nk = n - 1 - k + nk = (n-1)(k+1),$$

we obtain

$$\begin{aligned}\alpha(k)^2 + \frac{nk(n-1-k)}{(k+1)^2(n-1)^2} &= \frac{q(n-1)(k+1)}{(k+1)^2(n-1)^2} \\ &= \frac{q}{(k+1)(n-1)} \\ &= \alpha(k). \end{aligned} \tag{40}$$

Therefore,

$$\frac{1}{n}\sum_{i=1}^n \mathbb{E}\|u_i - \mu\|_2^2 = \alpha(k)\left(1 - \|\mu\|_2^2\right), \tag{41}$$

which proves Equation 26 in main paper.

We next relate Euclidean concentration around μ to angular concentration around $\hat{\mu}$. For any nonzero vectors x and y,

$$\left\|\frac{x}{\|x\|_2} - \frac{y}{\|y\|_2}\right\|_2 \le \frac{2\|x-y\|_2}{\|y\|_2}. \tag{42}$$

Indeed,

$$\begin{aligned}\left\|\frac{x}{\|x\|_2} - \frac{y}{\|y\|_2}\right\|_2 &\le \left\|\frac{x}{\|x\|_2} - \frac{x}{\|y\|_2}\right\|_2 + \left\|\frac{x-y}{\|y\|_2}\right\|_2 \\ &= \frac{|\|y\|_2 - \|x\|_2|}{\|y\|_2} + \frac{\|x-y\|_2}{\|y\|_2} \\ &\le \frac{2\|x-y\|_2}{\|y\|_2}, \end{aligned} \tag{43}$$

where the last inequality follows from the reverse triangle inequality

$$|\|x\|_2 - \|y\|_2| \le \|x - y\|_2.$$

Applying Equation 42 with $x = u_i$ and $y = \mu$ gives

$$\|\hat{u}_i - \hat{\mu}\|_2^2 \le \frac{4}{\|\mu\|_2^2}\|u_i - \mu\|_2^2. \tag{44}$$

Taking expectations and averaging over i,

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^n \mathbb{E}\|\hat{u}_i - \hat{\mu}\|_2^2 &\le \frac{4}{\|\mu\|_2^2}\frac{1}{n}\sum_{i=1}^n \mathbb{E}\|u_i - \mu\|_2^2 \\ &= \frac{4\alpha(k)\left(1 - \|\mu\|_2^2\right)}{\|\mu\|_2^2}. \end{aligned} \tag{45}$$

This proves Equation 27 in main paper.

For the pairwise cosine dispersion, note that for unit-normalized directions,

$$1 - \hat{u}_i^\top \hat{u}_j = \frac{1}{2}\|\hat{u}_i - \hat{u}_j\|_2^2. \tag{46}$$

Also,

$$\hat{u}_i - \hat{u}_j = (\hat{u}_i - \hat{\mu}) - (\hat{u}_j - \hat{\mu}).$$

Using

$$\|a - b\|_2^2 \le 2\|a\|_2^2 + 2\|b\|_2^2,$$

we obtain

$$\begin{aligned}1 - \hat{u}_i^\top \hat{u}_j &= \frac{1}{2}\|\hat{u}_i - \hat{u}_j\|_2^2 \\ &\le \|\hat{u}_i - \hat{\mu}\|_2^2 + \|\hat{u}_j - \hat{\mu}\|_2^2. \end{aligned} \tag{47}$$

Averaging Equation 47 over all ordered pairs $i \ne j$ gives

$$\begin{aligned}\mathbb{E}[\mathrm{MPD}_{\cos}(U)] &\le \frac{1}{n(n-1)}\sum_{i\ne j}\left[\mathbb{E}\|\hat{u}_i - \hat{\mu}\|_2^2 + \mathbb{E}\|\hat{u}_j - \hat{\mu}\|_2^2\right] \\ &= \frac{2}{n}\sum_{i=1}^n \mathbb{E}\|\hat{u}_i - \hat{\mu}\|_2^2 \\ &\le \frac{8\alpha(k)\left(1 - \|\mu\|_2^2\right)}{\|\mu\|_2^2}. \end{aligned} \tag{48}$$

Since cosine distance between two unit vectors lies in $[0, 2]$,

$$\mathrm{MPD}_{\cos}(U) \le 2.$$

Combining the two bounds therefore gives

$$\mathbb{E}[\mathrm{MPD}_{\cos}(U)] \le \min\left\{2,\ \frac{8\alpha(k)\left(1 - \|\mu\|_2^2\right)}{\|\mu\|_2^2}\right\}.$$

When $0 < \|\mu\|_2 < 1$, the factor

$$\frac{8(1 - \|\mu\|_2^2)}{\|\mu\|_2^2}$$

is strictly positive and independent of k. Since $\alpha(k)$ strictly decreases with k, the nontrivial analytic bound also strictly decreases.

If instead

$$\|\mu\|_2 = 1,$$

then

$$\left\| \frac{1}{n} \sum_{i=1}^{n} v_i \right\|_2 = \frac{1}{n} \sum_{i=1}^{n} \|v_i\|_2.$$

Equality in the triangle inequality for unit vectors requires all v_i to have the same direction. Thus the original space is already fully aligned, and the angular bound is identically zero.

Finally, at global pooling,

$$k = n - 1.$$

Each S_i then contains all vectors except v_i, so

$$\begin{aligned} u_i &= \frac{1}{n} \left(v_i + \sum_{j \neq i} v_j \right) \\ &= \frac{1}{n} \sum_{j=1}^{n} v_j \\ &= \mu. \end{aligned} \tag{49}$$

Since $\mu \neq 0$,

$$\hat{u}_i = \hat{\mu} \qquad \text{for all } i,$$

and hence

$$\mathrm{MPD}_{\cos}(U) = 0.$$

□